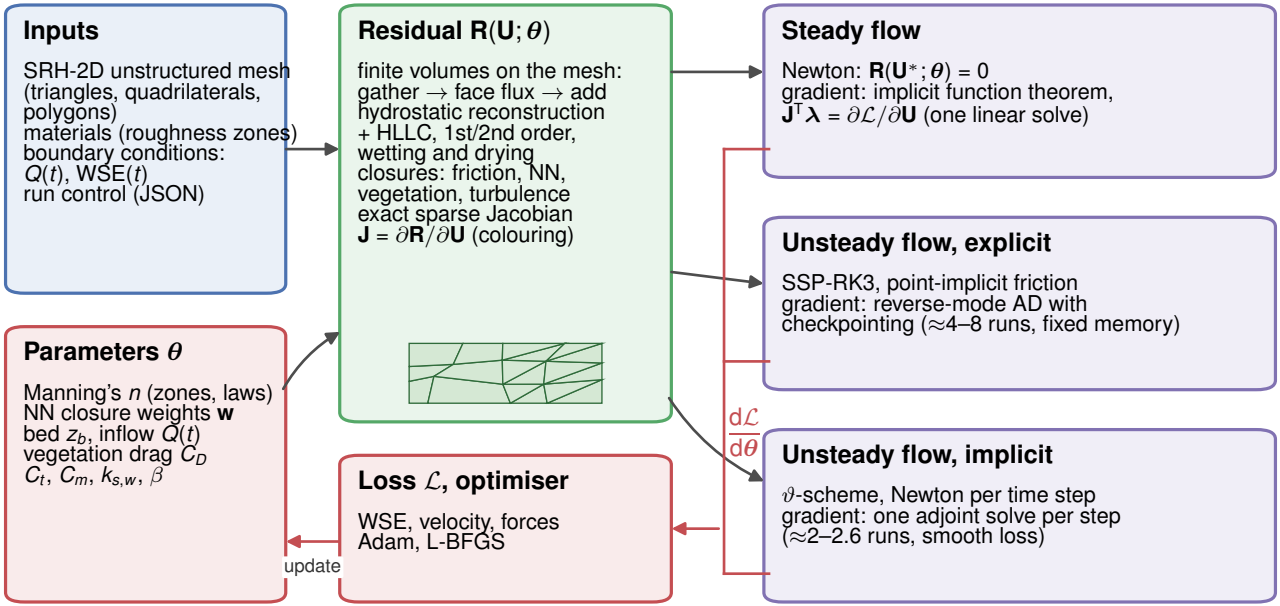

Inputs
SRH-2D unstructured mesh (triangles, quadrilaterals, polygons)
materials (roughness zones)
boundary conditions: $Q(t)$, WSE(t)
run control (JSON)
Residual $\mathbf{R}(\mathbf{U};\theta)$
finite volumes on the mesh: gather → face flux → add
hydrostatic reconstruction + HLLC, 1st/2nd order, wetting and drying
closures: friction, NN, vegetation, turbulence
exact sparse Jacobian $\mathbf{J} = \partial\mathbf{R}/\partial\mathbf{U}$ (colouring)
Steady flow
Newton: $\mathbf{R}(\mathbf{U}^*;\theta) = 0$
gradient: implicit function theorem, $\mathbf{J}^T\boldsymbol{\lambda} = \partial\mathcal{L}/\partial\mathbf{U}$ (one linear solve)
Unsteady flow, explicit
SSP-RK3, point-implicit friction
gradient: reverse-mode AD with checkpointing (≈4–8 runs, fixed memory)
Unsteady flow, implicit
ϑ-scheme, Newton per time step
gradient: one adjoint solve per step (≈2–2.6 runs, smooth loss)
$\frac{d\mathcal{L}}{d\theta}$
Parameters θ
Manning's n (zones, laws)
NN closure weights $\mathbf{w}$
bed z_b, inflow $Q(t)$
vegetation drag C_D
C_t, C_m, $k_{s,w}$, β
Loss $\mathcal{L}$, optimiser
WSE, velocity, forces
Adam, L-BFGS
update